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CPCS-214 Computer Architecture and Organization

01 - Computer Technology

Introduction

- Computers are product of the incredibly vibrant IT industry
- This unusual industry embraces innovation at a breath-taking rate
- In last 30 years, there have been a number of new computers whose introduction appeared to revolutionize the computing industry
- These revolutions were cut short only because someone else built an even better computer

Revolution

- Computers have led to a third revolution for civilization
 - Information revolution taking its place alongside agricultural and industrial revolutions
- Affected our everyday lives profoundly
- Changed ways in which search for new knowledge is carried out

Revolution

- New vein of scientific investigation
 - Computational scientists joining theoretical and experimental scientists in the exploration of new frontiers in
 - Astronomy
 - Biology
 - Chemistry
 - Physics
 - Others

Revolution

- Computer revolution continues
- Applications that were economically infeasible suddenly become practical
 - In the recent past, the following applications were “CS fiction”:
 - *Computers in automobiles*
 - *Cell phones*
 - *Human genome project*
 - *WWW*
 - *Search engines*

Impact

- Advances in this technology now affect almost every aspect of our society
- Hardware (HW) advances have allowed programmers to create useful software (SW)
 - Explains why computers are omnipresent

Impact

- Today's science fiction suggests tomorrow's killer applications
- Already on their way are
 - Glasses that augment reality
 - Cashless society
 - Cars that can drive themselves
 - ...

Classes of Computing Applications

- Common set of HW technologies is used in computers ranging from smart home appliances to cell phones to supercomputers
- Have different design requirements and employ the core HW technologies in different ways

Classes of Computing Applications

- Computers are used in three different classes of applications
 - PCs
 - Servers
 - Embedded computers

Personal Computers (PCs)

- A computer designed for use by an individual, usually incorporating a graphics display, a keyboard, and a mouse
- Best-known form of computing
- Emphasize delivery of good performance to single users at low cost and usually computing technologies
- Drove the evolution of many computing technologies, which is only about 35 years old

Servers

- A computer used for running larger programs for multiple users, often simultaneously
- Typically accessed only via a network
- Built from same technology as desktop computers
- Span the widest range in cost and capability
- Little more than desktop computer without screen or keyboard
- Cost a thousand dollars
- File storage, small business applications

Supercomputers

- A class of computers with the highest performance and cost
- Configured as servers
 - Set at another extreme of servers
- Consist of
 - tens of thousands of processors
 - many terabytes of memory

Supercomputers

- Cost tens to hundreds of millions
- Use
 - Massive and extensive computations
 - Weather forecasting
 - Oil exploration

Embedded Computers

- A computer inside another device used for running one predetermined application or collection of software
- Largest class
- Span widest range of applications and performance
- Include
 - Microprocessors (μ Ps) in your car
 - TV set
 - Networks of processors that control modern airplane or cargo ship

Embedded Computers

- Designed to run one application or one set of related applications integrated with HW and delivered as single system
- Despite large number, most users never really see that they are using a computer

Decimal and Binary Values and Names

Decimal term	Abbreviation	Value	Binary term	Abbreviation	Value	% Larger
kilobyte	KB	10^3	kibibyte	KiB	2^{10}	2%
megabyte	MB	10^6	mebibyte	MiB	2^{20}	5%
gigabyte	GB	10^9	gibibyte	GiB	2^{30}	7%
terabyte	TB	10^{12}	tebibyte	TiB	2^{40}	10%
petabyte	PB	10^{15}	pebibyte	PiB	2^{50}	13%
exabyte	EB	10^{18}	exbibyte	EiB	2^{60}	15%
zettabyte	ZB	10^{21}	zebibyte	ZiB	2^{70}	18%
yottabyte	YB	10^{24}	yobibyte	YiB	2^{80}	21%

FIGURE 1.1 The 2^x vs. 10^y bytes ambiguity was resolved by adding a binary notation for all the common size terms. In the last column we note how much larger the binary term is than its corresponding decimal term, which is compounded as we head down the chart. These prefixes work for bits as well as bytes, so *gigabit* (Gb) is 10^9 bits while *gibibits* (Gib) is 2^{30} bits.

PostPC Era

- Continuing march of technology brings about generational changes in computer HW that shake up the entire IT industry
 - We have undergone such a change
 - Replacing PC is the **personal mobile device (PMD)**

PMDs

- Small wireless devices to connect to Internet
- Battery operated
- Typically cost hundreds of dollars
- Software is installed by downloading apps

PMDs

- Like PCs
 - users can download SW (“apps”) to run on them
- Unlike PCs
 - no longer have keyboard and mouse
 - more likely to rely on a touch-sensitive screen or even speech input
- Today’s PMD is a smart phone or a tablet computer
- But tomorrow may include electronic glasses

PostPC Era

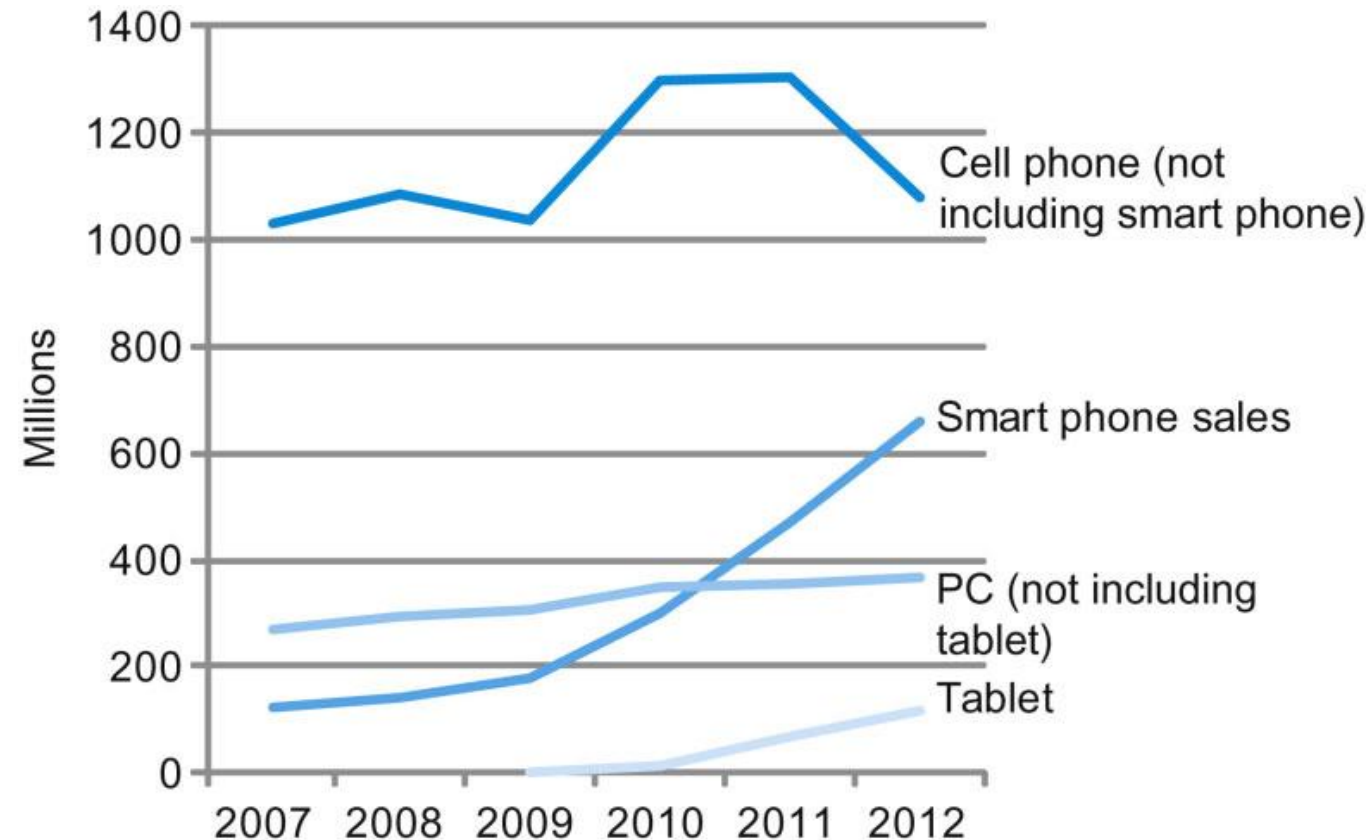


FIGURE 1.2 The number manufactured per year of tablets and smart phones, which reflect the PostPC era, versus personal computers and traditional cell phones. Smart phones represent the recent growth in the cell phone industry, and they passed PCs in 2011. Tablets are the fastest growing category, nearly doubling between 2011 and 2012. Recent PCs and traditional cell phone categories are relatively flat or declining.

Cloud Computing

- Large collections of servers that provide services over Internet
- Some providers rent dynamically varying numbers of servers as a utility
- Relies upon giant datacenters known as *Warehouse Scale Computers* (WSCs)
- Companies like Amazon and Google build WSCs containing 100,000 servers
 - Let companies rent portions of them
 - Can provide SW services to PMDs without having to build WSCs of their own

SaaS

- **Software as a Service** is deployed via cloud
 - Delivers SW and data as a service over Internet
 - Usually via a thin program such as a browser that runs on local client devices, instead of binary code that must be installed, and runs wholly on that device
- Examples include web search and social networking
- Today's SW developers will often have
 - Portion of their application that runs on PMD
 - Portion that runs in cloud